Alejandro Rosales

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Qualifications Summary

Computational Biologist/Data Engineer Intern with demonstrated expertise in driving efficiency, empowering transformation, and overseeing data optimization, automation, and agile methodologies.

Software / Data Engineering

- Ability to develop and maintain software applications and data pipelines to support various business needs.
- Capable of building data architectures for efficient storage, retrieval, and processing of large datasets.

Web Applications Programming

- Skilled in designing responsive and interactive web applications that meet user needs and business objectives.
- Well-versed in using artificial personal assistants; AWS, SQL, RESTful APIs, JavaScript, HTML/CSS, and Python.

Project Management

- Adept at spearheading projects from conception to completion, ensuring delivery within time and budget.
- Remarkable efficiency in monitoring project progress, tracking milestones, and managing project risks.

Key Strengths

- Instrumental in driving enhancements in current and established computing systems.
- Expert in leveraging cloud computing and programming languages to craft functional and dynamic solutions.

Career Experience

Brightseed, South San Francisco, CA **2022 – 2023**
Computational Biologist/Data Engineer Intern

Decreased data access time for Brightseed's proprietary information from span of weeks to just hours by directing integration of Neo4j graph database and internal API solutions as well as conducting comprehensive data analysis in collaboration with senior data engineer. Revamped setup efficiency for Brightseed's core AI algorithms, minimizing configuration time from days to mere minutes by constructing containerized environments through Docker. Applied Agile methodology, maintained code documentation, version control via Git, unit testing, and team repository management on GitHub.

- Achieved 10x enhancement in efficiency for generating core AI pipeline data packages at Brightseed by implementing automation techniques in cooperation with senior data engineer.
- Streamlined computation of 2.5B AI predictions and cut compute time from three weeks to three days by meticulous program profiling, time and resource optimization, and strategic implementation of parallel processing across entire Brightseed server.
- Increased core AI algorithm efficiency at Brightseed by 30% via autonomous creation and deployment of front-end and back-end components for internal website.
- Executed projects leveraging AWS cloud solutions, such as S3, EC2, ECS, ALB, VNC, and Lambda, while employing diverse range of programming languages and technologies, like Python, PHP, Angular, Docker, REST API, JavaScript, JSON, Typescript, jQuery, Ajax, SQL, NoSQL, MongoDB, and Neo4j.
- Presented three projects company wide, covering implications, use cases, and methodologies under director of computation.

The Coder School, Westerville, OH **2019 – 2021**
Code Coach

Developed students' programming skills with emphasis on runtime and space optimization, supporting objectives in website design, language acquisition, and machine learning through targeted coaching.

- Designed standardized Python curriculum for summer camps, selecting teaching materials and course content per school requirements.

Education

Master's Student in Computer Science | Focus in Quantum Machine Learning Ohio University, Athens, OH

- Relevant Coursework: Quantum Information Processing, Hardware for Deep Learning, Advanced Algorithms
- Research: Quantum Convolutional Neural Networks (QCNNs), Quantum Error Correction, Quantum Low-Density Parity-Check (qLDPC) Codes
- Related Papers: A. Rosales and A. Yadav, "Low-Overhead Error-Corrected QCNNs Using Bivariate Bicycle Codes," submitted to IEEE International Conference on Quantum Computing and Engineering (QCE '26), 2026. Paper ID: 643 (under review). Tutorial Journal Paper on QCNNs. MS Thesis

Bachelor of Science in Computer Science | Minor in Philosophy, Miami University, Oxford, OH

- Relevant Coursework: Computational Theory, Artificial Intelligence, Website Design, App Design
- Academic Affiliations: Curriculum Executive of Miami University Python Club, Miami University Programming Club